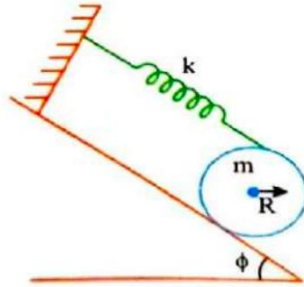
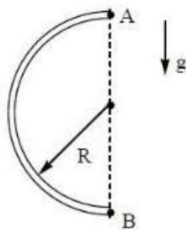


1. A uniform solid cylinder of mass m and radius R is in equilibrium on an inclined plane by the action of a light spring of stiffness k (gravity and reaction force acting on it). If the angle of inclination of the plane is ϕ , then angular frequency of small oscillations of the cylinder is (no slipping anywhere) ($m = 8\text{ kg}$, $K = 4\text{ N/m}$)



- A) 1 rad/sec ~~B) 2 rad/sec~~ C) 3 rad/sec D) 4 rad/sec
2. A particle is executing SHM on a straight line. A and B are two points at which its velocity is zero. It passes through a certain point P ($AP < PB$) at successive intervals of 0.5s and 1.5s with a speed of 3m/s. The maximum speed of the particle is
- A) 3 m/s B) 6 m/s C) $3\sqrt{3}\text{ m/s}$ ~~D) $3\sqrt{2}\text{ m/s}$~~
3. A particle moves with a simple harmonic motion in a straight line. In the first second starting from rest it travels a distance a and in the next second it travels a distance b in the same direction. The amplitude of the motion is
- A) $\frac{2a^2}{3b-a}$ B) $\frac{3a^2}{3a-b}$ ~~C) $\frac{2a^2}{3a-b}$~~ D) $\frac{3a^2}{3b-a}$
4. The end A of a thin uniform semi-circular wire of mass m and radius of curvature R is pivoted on a smooth hinge in uniform gravitational field and it is released from rest from the position shown (Diameter AB is along the vertical initially). Select the CORRECT statements from the following.



A) for small oscillations in its plane about its mean position, Its time period is

$$2\pi\sqrt{\frac{R}{2g}}$$

~~B)~~ for small oscillations in its plane about its mean position, Its time period

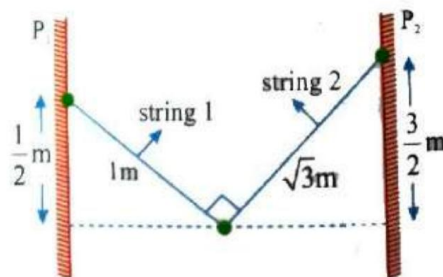
$$\text{is } 2\pi^2\sqrt{\frac{2R}{g(\pi^2 + 4)}}$$

~~C)~~ Angle made by the diametric line AB with vertical in equilibrium is $\tan^{-1}(2/\pi)$

~~D)~~ The maximum kinetic energy of wire in the subsequent motion after it is released

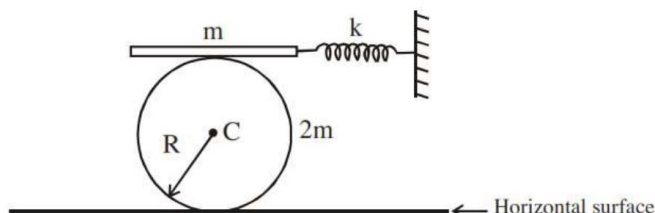
$$\text{from the position shown in the figure is } mgR\left(\sqrt{1+\frac{4}{\pi^2}}-1\right)$$

5. A small particle is tied by the help of two strings to vertical walls with the parameters as shown in the figure ($g = \pi^2$). The time period of the small oscillations of the particle if it is displaced perpendicular to the plane of the paper slightly is x sec, the value of x is



2

6. A uniform plank of mass m, free to move in horizontal direction, is placed on the top of a solid cylinder of mass 2m and radius R. The plank is attached with one end of a massless spring of force constant k and the other end of spring is fixed with a wall as shown in fig. The plank is slightly displaced towards spring and released. There is no slipping anywhere then



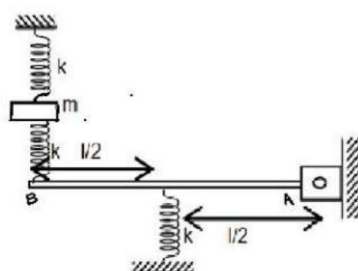
~~A)~~ Total energy of the system, when compression is spring is x and velocity of plank is v , is $\frac{7}{8}mv^2 + \frac{1}{2}kx^2$

B) Angular frequency of oscillation is $\sqrt{\frac{4k}{7m}}$

C) Time period of oscillation is $\pi\sqrt{\frac{7m}{4k}}$

~~D)~~ Total energy of the system, when compression is spring is x and velocity of plank is v , is $\frac{1}{2}kx^2 + \frac{11}{16}mv^2$

7. Figure shows a light rod AB of length 'l' which is smoothly pivoted at the end A as shown. What will be angular frequency (rad/sec) for small oscillation of block of mass 'm' in this arrangement shown. Neglect the mass of the rod. $\left(\frac{k}{m} = \frac{10}{3} \text{ SI unit}\right)$



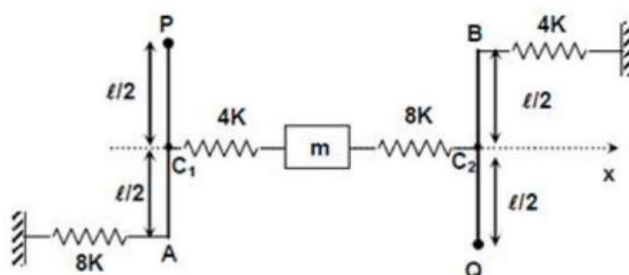
~~A)~~ 2

B) 3

C) 7

D) 8

8. In the figure shown, arrangement is placed on horizontal frictionless plane. The mass of rod PA and QB are negligible and they are free to rotate about hinge points P and Q respectively in the horizontal plane. The block of mass 'm' is attached to springs. Springs are light and their stiffness are shown in the figure. The block is slightly displaced along x-axis then released. Then the time period of oscillation of the block is _____.



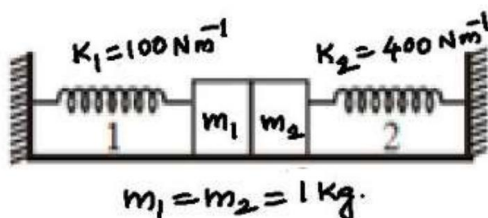
~~A)~~ $\frac{3\pi}{2}\sqrt{\frac{m}{5k}}$

B) $2\pi\sqrt{\frac{13m}{k}}$

C) $4\pi\sqrt{\frac{m}{k}}$

D) $\pi\sqrt{\frac{m}{2k}}$

9. In the given figure m_1 & m_2 are attached to springs on left and right respectively and kept on smooth level ground. The left block is pushed to the left to compress spring 1 by 10cm and right block is pushed to right to compress spring 2 by 10 cm. Both the blocks are released simultaneously. Springs are initially relaxed as shown in the diagram.

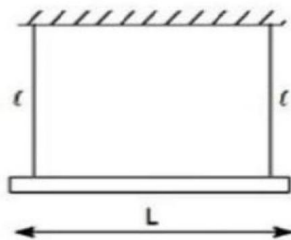


- A) They will collide at time when both springs come to the natural length
 B) They will collide at a point which is 5cm to left of position shown
 C) They will first collide at $t = \frac{\pi}{30} \text{ sec}$ D) They will first collide at $t = \frac{\pi}{10} \text{ sec}$
- 10 A rod of lengths L is supported by two ideal strings of length ℓ such that the system hangs in a vertical plane.

Case 1 : Rod is kept horizontal, displaced slightly perpendicular to the plane and allowed by oscillate.

Case 2 : Rod is given a small twist about central axis and then allowed to oscillate.

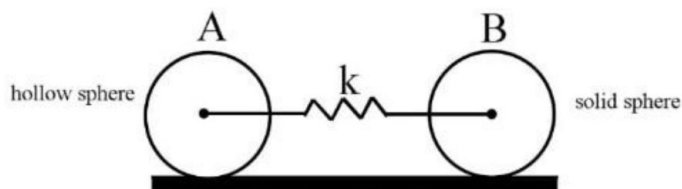
Let T_1, T_2 be the periods of oscillations in the two respective cases.



- A) $T_1 = 2\pi \sqrt{\frac{\ell}{g}}$ B) $T_1 = 2\pi \sqrt{\frac{\ell}{2g}}$ C) $T_2 = 2\pi \sqrt{\frac{L}{6g}}$ D) $T_2 = 2\pi \sqrt{\frac{\ell}{3g}}$

ans=ad

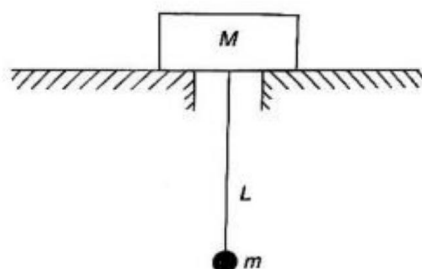
- 11 Two spheres A and B of the same mass m and the same radius are placed on a rough horizontal surface. A is a thin uniform hollow sphere and B is uniform solid sphere. A and B can roll without sliding on the floor. They are also tied centrally to light spring of spring constant k as shown figure. A and B are released when the extension in the spring is x_0 . The amplitude of SHM of the sphere B is $\frac{5N}{46} x_0$. Then find N



ans 5

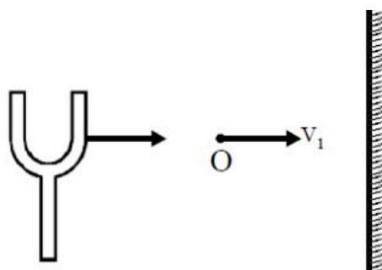
- 12 A block of mass M rests on a smooth horizontal table. There is a small gap in the table under the block through which a pendulum has been attached to the block. The mass of the bob of pendulum is ' m ' and length of the pendulum is ' L ', The pendulum is set into small oscillations in the vertical plane the figure. The table does not interfere with the motion of string, then the time period oscillations of pendulum is found out to be

$2\pi\sqrt{\frac{3L}{xg}}$, find ' x ' ? (Given $M=3m$)



ans=4

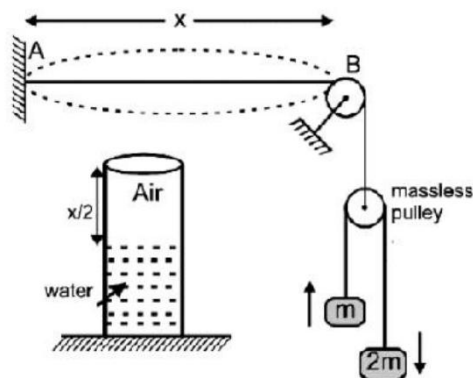
- 13 There is a source of sound (a tuning fork) moving towards a reflecting wall with a speed of 30m/s. The velocity of sound in air is 330m/s and the frequency of the tuning fork is 600Hz. An observer in between the source and the wall is moving with a velocity v_1 towards the wall. The beat frequency heard by the observer is 4Hz. If the tuning fork is now filed the beat frequency as heard by the observer is 8Hz. By what factor the frequency of tuning fork increases?



ans=2

- 14 AB is a wire vibrating in its fundamental mode. The wire AB is kept under tension by connecting it to two blocks of masses ' m ' and ' $2m$ ' as shown in the figure. The wire is in resonance with resonance tube in which air column is also vibrating in fundamental mode. Speed of sound in air is 400m/s and the linear mass density of the material of the wire is $10^{-4}kg/m$ and $g=10m/s^2$, then the value of ' m ' is $\frac{\beta}{10}kg$. Find the value of β ?
(Neglect the mass of the wire when compared to the masses of blocks)

(Neglect the mass of the wire when compared to the masses of blocks)



ans=6

- 15 A triangular wave pulse on a taut string travels in positive x -direction with speed ' v '. The tension in the string is ' F ' and linear density of the string is ' μ '. At $t=0$ the shape of the pulse is given by

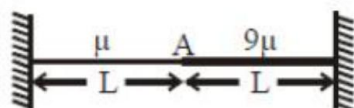
$$y(x,0) = \begin{cases} 0 & \text{if } x < -L \\ \frac{h(L+x)}{L} & \text{for } -L < x < 0 \\ \frac{h(L-x)}{L} & \text{for } 0 < x < L \\ 0 & \text{for } x > L \end{cases}$$

Choose the **INCORRECT** statement(s)

- A) Magnitude of the instantaneous power is zero for $-L < x - vt < 0$
- B) Magnitude of the instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for $(x - vt) < -L$
- C) Magnitude of the instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for $(x - vt) > L$
- D) Magnitude of the instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for $0 < (x - vt) < L$

ans=abc

- 16 A light string is tied at one end to a fixed support and to a heavy string of equal length L at the other end as shown in the figure. Mass per unit length of the strings are μ and 9μ respectively and the tension in the string is T . Then the possible values of frequencies such that point A is an antinode are



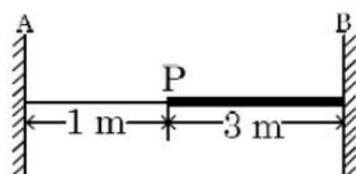
- A) $\frac{1}{4L}\sqrt{\frac{T}{\mu}}$ B) $\frac{3}{4L}\sqrt{\frac{T}{\mu}}$ C) $\frac{3}{2L}\sqrt{\frac{T}{\mu}}$ D) $\frac{7}{6L}\sqrt{\frac{T}{\mu}}$

ans=ab

- 17** A point sound source is situated in a medium of bulk modulus $1.6 \times 10^5 \text{ Nm}^{-2}$. An observer standing at a distance 10m from the source writes down the equation of the wave as $y = a \sin(15\pi x - 6000\pi t)$, here y and x are in meters and t is in seconds. The maximum pressure amplitude received by the observer's ear is 24π Pa, Then
- A) The density of the medium is 1 kg/m^3
- B) The displacement amplitude A of the waves as received by the observer is $10 \mu\text{m}$.
- C) Power of the sound source is $288\pi^3$ Watt
- D) Power of the sound source is $188\pi^3$ Watt

abc

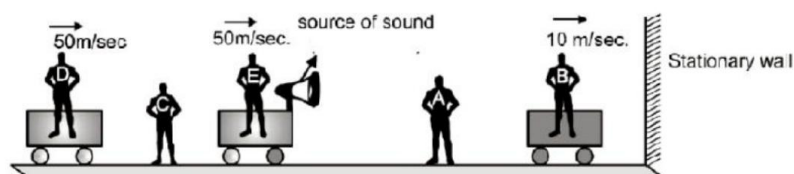
- 18** A light string of length 1m and mass 10g is tied at one end to a fixed support and the other end to a heavy string of length 3m and mass 270g as shown in the figure. Both the strings are kept at a constant tension of 16N. The whole system is excited by a source of transverse vibrations of varying frequencies, such that the light string supports one or more loops. Then pick out the correct statement(s) of the following.



- A) The point P corresponds to a node, the minimum number of loops formed along the heavy string will be 9
- B) The point P corresponds to an antinode for the minimum frequency of vibration of 30Hz
- C) The point P corresponds to a node if the possible frequencies of vibration are integral multiples of 10Hz
- D) The point P corresponds to an antinode, the minimum number of loops formed in the heavy string must be 14

abd

- 19** A source of sound emitting frequency 300Hz moving towards a stationary reflecting wall with a speed 50m/s. There are five observers A,B,C,D and E as shown in the figure. Speed of sound is 350m/s. Then



- A) The beat frequency as heard by observer A is ZERO
- B) The beat frequency as heard by observer B is 20Hz
- C) The beat frequency as heard by observer E is 20Hz
- D) The beat frequency as heard by observer C is 20Hz

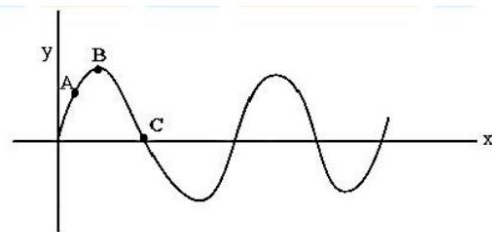
ab

- 20** There are two narrow pipes A and B of equal length l . Pipe A is filled with Helium gas and pipe B is filled with hypothetical diatomic gas whose molar mass is $7/6$ times the molar mass of hydrogen. Both the gases are at the same temperatures. Then

- A) Pipe A can resonate with pipe B if both pipes A and B are open pipes
- B) Pipe A can resonate with pipe B if pipe A is closed and pipe B is open
- C) Pipe A can resonate with pipe B if pipe A is open and pipe B is closed
- D) Pipe A can resonate with pipe B if both pipes A and B are closed

ac

- 21** A plane progressive transverse wave is passing through medium along the positive x-axis at the given instant of time. The nature of the displacement – position (y vs x) graph of particles of medium at that instant is as shown in figure. Which of the following statements are correct . (A,B,C are three particles of the vibrating string)



- (1) Elastic potential energy per unit length of the string at the point B is more than that at points A and C
- (2) Elastic potential energy per unit length of the string at the point C is more than that at points A and B
- (3) velocity of the particle at A is along negative y-axis.
- (4) Velocity of the particle at C is along positive y-axis.

- A) Only 2,3,4 are correct
- B) Only 1,3,4 are correct
- C) Only 1,3 are correct
- D) Only 1,4 are correct

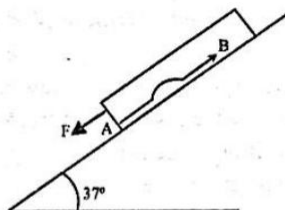
ans=a

- 22** A source of sound with natural frequency f_0 moves uniformly along a straight line separated from a stationary observer by a distance ' l ' ($\perp^{\text{ar}}$ distance). The velocity of source is equal to $\eta (< 1)$ times of velocity of sound. Find the frequency of sound received by the observer at the moment when the source gets closest to him?

A) f_0 B) $\frac{f_0}{1-\eta^2}$ C) $\frac{f_0}{1+\eta^2}$ D) $\frac{f_0}{1-\eta}$

ans=b

- 23** A box of mass $14m$ with smooth base is sliding on an inclined plane of inclination θ under the influence of a force F . A string of mass m length l is attached at A. The box is released from rest and a transverse pulse is produced at other end 'B'. Mark the correct statement(s) for the pulse.



- A) If $F = 0$ the pulse will remain stationary
 B) If $F = 0$ the pulse will not reach to end A
 C) $F = 9mg$ the time taken by the pulse in moving from B to A will be $2\sqrt{\frac{5l}{3g}}$
 D) If $F = 18\text{ mg}$ the time taken by the pulse in moving from B to A will be $2\sqrt{\frac{5l}{3g}}$

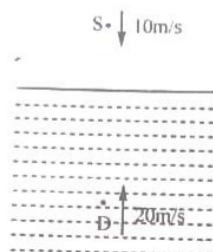
ans=bc

- 24** A transverse harmonic wave is propagating along a taut string. Tension in the string is 50 N and its linear mass density is 0.02 kg/m . The string is driven by a 80 Hz oscillator tied to one end, oscillating with amplitude of 1 mm . The other end of string is terminated so that all the wave energy is absorbed and there is no reflection. From this given data, choose the correct option(s)

- A) Power of the oscillator is 0.126 w
 B) If tension in the string is quadrupled and power of oscillator remains same. Then new amplitude of oscillation of string is $\frac{1}{\sqrt{2}}\text{ mm}$
 C) Average energy of the wave on a 1 m long segment of the string is 2.5 mJ
 D) Power of the oscillator is 0.252 w

ans=abc

- 25** A person riding on a source 'S' of sound (frequency 48Hz) moves in a medium (in which velocity of sound is 310 m/s) with speed 10 m/s as shown towards a liquid reservoir (in which velocity of sound is 1240 m/s). A detector 'D' in the reservoir moves towards the free surface of the reservoir (along line SD) at a speed 20m/s. The sound produced by 'S' gets reflected and transmitted at the free surface. Mark the INCORRECT statement(s) among the following, assuming the liquid in the reservoir stays still and calm.



- A) Wavelength of sound received by detector D is 25m
 B) Beat frequency registered by the person is 1.6Hz
 C) Frequency of sound detected by detector D is 56Hz
 D) A compression pulse incident on the free surface due to 'S' is reflected as a rarefaction pulse

ans=bcd

- 26** Mark the correct statement(s)

- A) When a sound wave strikes a wall, the compression pulse is reflected as compression pulse
 B) When a sound wave strikes a wall, the compression pulse is reflected as rarefaction pulse
 C) When a sound wave is coming out after passing through a narrow pipe, then reflection would be there at open end.
 D) When a sound wave is coming out after passing through a narrow pipe, then compression pulse reflected as rare fraction pulse.

ans=acd

- 27** A glass tube of 1.0 m length is filled with water. The water is drained out very slowly from the tube through a hole in the bottom while a vibrating 500 Hz tuning fork is held near the open upper end of the tube. If the speed of sound is 320 ms^{-1} find the number of resonances that can be obtained.

ans=3

- 28** A sound wave of angular frequency ω travels with a speed v in a medium of density ρ and bulk modulus B . Let k be the propagation constant. If p and A are the pressure amplitude and displacement amplitude respectively, then the intensity of sound wave is

- A) $\frac{1}{2} \omega B k A^2$ B) $\frac{v p^2}{2B}$ C) $\frac{p^2}{2 \rho v}$ D) $\frac{p^2}{2 \sqrt{\rho B}}$

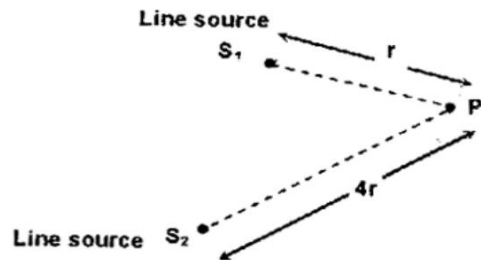
ans=abcd

- 29** The superposition of two waves travelling in opposite directions given by the equations: $y_1 = a \sin(\omega t - kx)$ and $y_2 = 2a \sin(\omega t + kx)$ Takes place in a homogeneous string of cross section area s and density ρ . The resulting pattern can be thought as a superposition of a standing wave and progressive wave travelling left. The total mechanical energy between two successive antinodes is:

A) $\frac{3\pi s \rho \omega^2 a^2}{2k}$ B) $\frac{\pi s \rho \omega^2 a^2}{2k}$ C) $\frac{5\pi s \rho \omega^2 a^2}{2k}$ D) $\frac{\pi s \rho \omega^2 a^2}{k}$

ans=c

- 30** Intensity due to a line source S_1 at a distance of r is I_0 and intensity due to other line source S_2 at a distance of r is $4I_0$ and frequency of both source are equal. If interference occurs at a point P as shown in the figure and wavelength of both sources is r , find the resultant intensity (in watt/m²) at P. (where $I_0 = 1$ watt/m²)



ans=4